Assesment-1(Operating System)

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Q.1

Aim : To implement shell commands

- ls -la — List all files (including hidden) with details

```
drwxr-x---+  77 niteshpandey  staff   2464 22 Aug 11:52 .
drwxr-xr-x    5 root          admin    160 20 Jul 01:24 ..
drwxr-xr-x    4 niteshpandey  staff    128 27 Mar 06:46 .anaconda
-rw-r--r--    1 niteshpandey  staff    447 27 Mar 06:43 .bash_profile
-rw-------    1 niteshpandey  staff      3  6 Aug 20:07 .bashrc.save
drwxr-xr-x    5 niteshpandey  staff    160 19 Aug 15:13 .cache
drwxr-xr-x@   3 niteshpandey  staff     96  8 Jul 17:23 .cagent
-r--------    1 niteshpandey  staff      7  3 Jan  2026 .CFUserTextEncoding
drwxr-xr-x   20 niteshpandey  staff    640 22 Aug 11:50 .claude
-rw-------@   1 niteshpandey  staff  48673 22 Aug 11:50 .claude.json
drwxr-xr-x    5 niteshpandey  staff    160  1 Apr 07:29 .conda
drwxr-xr-x@   6 niteshpandey  staff    192 18 Jul 15:51 .config
drwxr-xr-x    3 niteshpandey  staff     96 27 Mar 06:47 .continuum
drwx------@   5 niteshpandey  staff    160 26 Jul 18:14 .copilot
drwxr-xr-x@  11 niteshpandey  staff    352  2 Aug 07:24 .cursor
drwxr-xr-x@  12 niteshpandey  staff    384  8 Jul 22:19 .docker
-rw-------    1 niteshpandey  staff      0 22 Jan  2026 .docker-wine.Xkey
-rw-r--r--@   1 niteshpandey  staff  18436 19 Aug 21:16 .DS_Store
-rw-r--r--@   1 niteshpandey  staff     62 11 Jan  2026 .gitconfig
drwxr-xr-x    3 niteshpandey  staff     96 27 Mar 06:49 .ipynb_checkpoints
drwxr-xr-x    3 niteshpandey  staff     96 27 Mar 06:49 .ipython
drwxr-xr-x    4 niteshpandey  staff    128 27 Mar 06:51 .jupyter
-rw-------@   1 niteshpandey  staff     20  9 Feb  2026 .lesshst
drwxr-xr-x@   5 niteshpandey  staff    160 18 Aug 07:28 .local
drwxr-xr-x@   3 niteshpandey  staff     96  7 Feb  2026 .matplotlib
-rw-------    1 niteshpandey  staff    871 25 Jul 05:43 .mysql_history
drwxr-xr-x@   5 niteshpandey  staff    160  3 Aug 17:24 .mysqlsh-wb
drwxr-xr-x@   6 niteshpandey  staff    192  9 Feb  2026 .npm
drwxr-xr-x@   8 niteshpandey  staff    256 26 May 13:57 .ollama
-rw-------@   1 niteshpandey  staff     64  7 Feb  2026 .python_history
-rw-r--r--@   1 niteshpandey  staff   7641 17 Apr 06:12 .RData
-rw-r--r--@   1 niteshpandey  staff   3103 18 Apr 22:08 .Rhistory
-rw-r--r--    1 niteshpandey  staff    281 27 Mar 06:43 .tcshrc
drwx------+  14 niteshpandey  staff    448 17 Aug 21:21 .Trash
-rw-------    1 niteshpandey  staff   8157 19 Aug 21:25 .viminfo
drwxr-xr-x@   5 niteshpandey  staff    160  5 Jan  2026 .vscode
drwxr-xr-x@   3 niteshpandey  staff     96  8 Jun 03:50 .vscode-shared
-rw-------    1 niteshpandey  staff     70 22 Jan  2026 .Xauthority
-rw-r--r--    1 niteshpandey  staff    635 27 Mar 06:43 .xonshrc
-rw-r--r--    1 niteshpandey  staff    252  4 Feb  2026 .zprofile
-rw-r--r--    1 niteshpandey  staff     86  5 Jan  2026 .zprofile.pysave
-rw-------@   1 niteshpandey  staff  36813 22 Aug 10:25 .zsh_history
drwx------   24 niteshpandey  staff    768 22 Aug 11:52 .zsh_sessions
-rw-r--r--    1 niteshpandey  staff    545 18 Aug 07:30 .zshrc
-rw-r--r--    1 niteshpandey  staff   1422 27 Mar 06:55 100daysML.ipynb
drwxr-xr-x    3 niteshpandey  staff     96 18 Aug 07:33 Applications
-rwxr-xr-x    1 niteshpandey  staff  33800 18 Aug 14:50 checksum
-rw-r--r--    1 niteshpandey  staff   2896 18 Aug 14:49 cheksum.c
-rwxr-xr-x    1 niteshpandey  staff  43320  5 Aug 21:14 compiler
-rw-r--r--    1 niteshpandey  staff    423  5 Aug 21:13 compiler.cpp
-rw-r--r--    1 niteshpandey  staff   1264 19 Aug 21:15 compiler1.cpp
-rwxr-xr-x    1 niteshpandey  staff  40408  5 Aug 21:26 compiler2
-rw-r--r--    1 niteshpandey  staff    422  5 Aug 21:25 compiler2.cpp
drwx------+  48 niteshpandey  staff   1536 20 Aug 17:56 Desktop
drwx------+   5 niteshpandey  staff    160 30 Mar 07:50 Documents
```

- pwd — Print current working directory

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % pwd
/Users/niteshpandey
```

- cd /path/to/dir — Change directory

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % cd desktop
(base) niteshpandey@Niteshs-MacBook-Pro desktop %
```

- cp file1.txt file2.txt — Copy a file

```
Last login: Sat Aug 22 11:52:15 on ttys000
(base) niteshpandey@Niteshs-MacBook-Pro ~ % touch file.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ % touch file1.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ % cp file1.txt file2.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- mv old.txt new.txt — Move/rename a file

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % mv file.txt new.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- rm file.txt — Remove a file

```
rm: file.txt: No such file or directory
(base) niteshpandey@Niteshs-MacBook-Pro ~ % touch nitesh.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ % rm nitesh.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- mkdir new_folder — Create a directory

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % mkdir demo
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- rmdir empty_folder — Remove an empty directory

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % mkdir demo
(base) niteshpandey@Niteshs-MacBook-Pro ~ % rmdir demo
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- cat file.txt — Print file contents

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % echo "Hello, this is my first text file." > file.txt
(base) niteshpandey@Niteshs-MacBook-Pro ~ % cat file.txt
Hello, this is my first text file.
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- grep "pattern" file.txt — Search for text in a file

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % grep "this" file.txt
Hello, this is my first text file.
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- find . -name "*.txt" — Find files by name pattern

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % find . -name "*.txt"
./file2.txt
./file.txt
./.cursor/projects/Users-niteshpandey-Desktop-test-fin-FinTrack/agent-tools/13f796af-b4fc-4de6-b6f3-79a610e906dc.txt
./.cursor/projects/Users-niteshpandey-Desktop-test-fin-FinTrack/agent-transcripts/ee037802-4086-46a2-894b-11d63d7f749d.txt
./.cursor/projects/Users-niteshpandey-Desktop-ppppp-FinTrack/agent-transcripts/cb87e896-dae9-4aac-97bc-2eb0ac036188.txt
./.cursor/extensions/anysphere.remote-ssh-1.1.13/dist/main.js.LICENSE.txt
./.cursor/extensions/anysphere.remote-ssh-1.1.13/LICENSE.txt
./.cursor/extensions/anysphere.remote-containers-1.0.38/dist/@devcontainers/cli/dist/spec-node/devContainersSpecCLI.js.LICENSE.txt
./.cursor/extensions/anysphere.remote-containers-1.0.38/dist/main.js.LICENSE.txt
./.cursor/extensions/anysphere.remote-containers-1.0.38/LICENSE.txt
./file1.txt
./fintrack/node_modules/jsesc/LICENSE-MIT.txt
./fintrack/node_modules/next/dist/bundle-analyzer/__next._tree.txt
./fintrack/node_modules/next/dist/bundle-analyzer/__next._head.txt
./fintrack/node_modules/next/dist/bundle-analyzer/__next._full.txt
./fintrack/node_modules/next/dist/bundle-analyzer/__next._index.txt
```

- ps aux — List running processes

```
chmod: script.sh: No such file or directory
(base) niteshpandey@Niteshs-MacBook-Pro ~ % ps aux
USER               PID  %CPU %MEM      VSZ    RSS   TT  STAT STARTED      TIME COMMAND
niteshpandey      1206  12.6  0.2 435458208  59104   ??  S    20Jul26  88:15.51 /usr/libexec/duetexpertd
_windowserver      584   6.4  0.7 441672672 186048   ??  Ss   20Jul26 1021:55.41 /System/Library/PrivateFr
niteshpandey     98084   2.9  3.0 509741248 757760   ??  Ss   Sun12PM  41:32.42 /System/Library/Framework
_locationd         559   2.8  0.1 435427312  24768   ??  Ss   20Jul26  50:31.04 /usr/libexec/locationd
niteshpandey     54658   1.9  4.4 509207760 1110544   ??  Ss   12:16PM   1:59.33 /System/Library/Framework
niteshpandey       960   1.8  0.1 435374448  13264   ??  S    20Jul26   6:27.79 /System/Library/PrivateFr
niteshpandey     10458   1.8  0.2 440193840  59760   ??  S    26Jul26  44:41.43 /System/Applications/Note
niteshpandey     44236   1.3  1.9 443789392 490512   ??  S    30Jul26  61:12.52 /Applications/Microsoft W
root               552   1.0  0.1 435378544  17424   ??  Ss   20Jul26  11:32.12 /usr/libexec/opendirector
niteshpandey       969   1.0  0.1 435388368  29872   ??  S    20Jul26   2:06.57 /usr/libexec/routined LAU
niteshpandey     54633   0.9  0.5 435874080 130144   ??  S    12:16PM   0:06.07 /System/Applications/Util
niteshpandey       961   0.7  0.1 435456624  20800   ??  S    20Jul26   1:00.89 /usr/libexec/linkd
niteshpandey     96544   0.6  0.6 444707280 148032   ??  Rs   Sun11AM  52:17.56 /System/Library/Framework
niteshpandey       943   0.5  0.1 435379376  13760   ??  S    20Jul26  20:54.38 /usr/libexec/trustd --age
niteshpandey     96536   0.5  1.1 445939344 267104   ??  S    Sun11AM  39:00.19 /System/Volumes/Preboot/C
niteshpandey     96543   0.4  0.3 440226224  86976   ??  Ss   Sun11AM  14:53.88 /System/Library/Framework
_driverkit         683   0.3  0.2 435306560  43920   ??  Ss   20Jul26 164:13.95 /System/Library/DriverExt
_networkd          632   0.2  0.1 435390000  16128   ??  Ss   20Jul26  20:32.81 /usr/libexec/symptomsd
niteshpandey       903   0.2  0.0 435338272   5664   ??  S    20Jul26   5:52.12 /usr/sbin/distnoted agent
root                 1   0.1  0.1 435246272  24864   ??  Ss   20Jul26 125:13.82 /sbin/launchd
root               554   0.1  0.1 435374768  14048   ??  Ss   20Jul26   4:57.06 /System/Library/CoreServi
niteshpandey     28238   0.1  4.9 510057184 1240736   ??  Ss   Tue06AM  21:34.87 /System/Library/Framework
niteshpandey     24076   0.1  2.7 509557104 675744   ??  Ss   Thu04PM   4:30.84 /System/Library/Framework
root               551   0.1  0.2 435427296  47824   ??  Ss   20Jul26   0:13.72 /usr/libexec/xprotectd
niteshpandey     52585   0.1  0.5 436593504 132160   ??  SN   11:46AM   0:08.22 /Library/Frameworks/Pytho
root               614   0.1  0.0 435344144   6192   ??  Ss   20Jul26   0:35.41 /System/Library/PrivateFr
root               577   0.1  0.0 435339120   4704   ??  Ss   20Jul26   7:42.01 /usr/sbin/notifyd
niteshpandey     93161   0.1  2.6 509926880 661584   ??  Ss   Tue04PM   8:33.30 /System/Library/Framework
niteshpandey      1405   0.1  0.0 435528944  12192   ??  S    20Jul26   3:47.16 /System/Library/CoreServi
root               579   0.1  0.0 435374336   8320   ??  Ss   20Jul26  35:48.71 /usr/libexec/corebrightne
root               511   0.1  0.2 435472160  49488   ??  Ss   20Jul26  49:57.23 /usr/libexec/logd
_distnote          566   0.1  0.0 435338208   3088   ??  Ss   20Jul26   3:45.57 /usr/sbin/distnoted daemo
```

- df -h — Show disk usage (human-readable)

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % df -h
Filesystem        Size    Used   Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1   460Gi    12Gi   300Gi     4%    459k  3.1G    0%   /
devfs            355Ki   355Ki     0Bi   100%    1.2k     0  100%   /dev
/dev/disk3s6     460Gi   5.0Gi   300Gi     2%       5  3.1G    0%   /System/Volumes/VM
/dev/disk3s2     460Gi   8.4Gi   300Gi     3%    1.4k  3.1G    0%   /System/Volumes/Preboot
/dev/disk3s4     460Gi   3.7Mi   300Gi     1%      66  3.1G    0%   /System/Volumes/Update
/dev/disk1s2     500Mi   6.0Mi   481Mi     2%       1  4.9M    0%   /System/Volumes/xarts
/dev/disk1s1     500Mi   5.6Mi   481Mi     2%      32  4.9M    0%   /System/Volumes/iSCPreboot
/dev/disk1s3     500Mi   2.5Mi   481Mi     1%      84  4.9M    0%   /System/Volumes/Hardware
/dev/disk3s5     460Gi   134Gi   300Gi    31%    1.9M  3.1G    0%   /System/Volumes/Data
map auto_home      0Bi     0Bi     0Bi   100%       0     0     -   /System/Volumes/Data/home
/dev/disk4s1     1.2Gi   875Mi   317Mi    74%     21k  4.3G    0%   /Volumes/Cursor Installer
/dev/disk5s1     1.1Gi   815Mi   290Mi    74%    3.3k  4.3G    0%   /Volumes/Claude
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

- du -sh folder/ — Show size of a folder

```
(base) niteshpandey@Niteshs-MacBook-Pro desktop % du -sh RAG_PROBLEM_1
1.7G    RAG_PROBLEM_1
(base) niteshpandey@Niteshs-MacBook-Pro desktop %
```

- history — Show command history

```
(base) niteshpandey@Niteshs-MacBook-Pro desktop % history
 1007  touch nitesh.txt
 1008  rm nitesh.txt
 1009  mkdir demo
 1010  rmdir demo
 1011  cat file1.txt
 1012  echo "Hello, this is my first text file." > file.txt
 1013  cat file.txt
 1014  grep file.txt
 1015  grep "this" file.txt
 1016  find .-file"*.txt"
 1017  find . -name "*.txt"
 1018  chmod +x script.sh
 1019  ps aux
 1020  df -h
 1021  cd desktop
 1022  du -sh RAG_PROBLEM_1
(base) niteshpandey@Niteshs-MacBook-Pro desktop %
```

Q.2

Aim : create a child process using fork() system call create following process.

i. Orphan process
ii. Zombie process

Code and Output:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>


int main() {


  pid_t pid = fork();


  if (pid < 0) {

    printf("Fork failed\n");

    return 1;

  }


  if (pid == 0) {

    // Child process

    printf("Child process started\n");

    printf("Child PID: %d\n", getpid());

    printf("Parent PID: %d\n", getppid());


    sleep(10);
```

```
        printf("\nAfter parent terminates:\n");
        printf("Child PID: %d\n", getpid());
        printf("New Parent PID: %d\n", getppid());
    }

    else {
        // Parent process
        printf("Parent process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        printf("Parent terminating...\n");
        exit(0);
    }

    return 0;
}
```

Output.

```
[(base) niteshpandey@Niteshs-MacBook-Pro ~ % g++ orphan.cpp -o orphan                ]
[(base) niteshpandey@Niteshs-MacBook-Pro ~ % ./orphan                                ]
Parent process
Parent PID: 54950
Child PID: 54954
Parent terminating...
Child process started
Child PID: 54954
Parent PID: 54950
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
After parent terminates:
Child PID: 54954
New Parent PID: 1
[212121                                                                              ]
```

2.Zombie process

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {

   pid_t pid = fork();

   if (pid < 0) {
      printf("Fork failed\n");
      return 1;
   }

   if (pid == 0) {
      // Child
      printf("Child process\n");
      printf("Child PID: %d\n", getpid());

      printf("Child terminating...\n");
      exit(0);
   }

   else {
```

```
    // Parent
    printf("Parent process\n");
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    printf("Parent sleeping...\n");

    sleep(20);

    printf("Parent terminating...\n");
  }

  return 0;
}
```

Output:

```
(base) niteshpandey@Niteshs-MacBook-Pro ~ % vi zombie.cpp
(base) niteshpandey@Niteshs-MacBook-Pro ~ % g++ zombie.cpp -o zombie
(base) niteshpandey@Niteshs-MacBook-Pro ~ % ./zombie
Parent process
Parent PID: 54968
Child PID: 54969
Parent sleeping...
Child process
Child PID: 54969
Child terminating...
Parent terminating...
(base) niteshpandey@Niteshs-MacBook-Pro ~ %
```

Q.3 Implement CPU Scheduling Algorithm

1. FCFS

2.SJF

3. Round Robin

4.Priority

Aim: to implement CPU scheduling algorithms

1.FCFS

Code.

```
#include <stdio.h>


int main() {
   int n;
   int at[20], bt[20], ct[20], tat[20], wt[20];
   int i, time = 0;
   float avg_wt = 0, avg_tat = 0;


   printf("Enter number of processes: ");
   scanf("%d", &n);


   for (i = 0; i < n; i++) {
      printf("\nEnter Arrival Time for P%d: ", i + 1);
      scanf("%d", &at[i]);


      printf("Enter Burst Time for P%d: ", i + 1);
      scanf("%d", &bt[i]);
```

```
}

// Sort according to Arrival Time
for (i = 0; i < n - 1; i++) {
   for (int j = 0; j < n - i - 1; j++) {
      if (at[j] > at[j + 1]) {
         int temp;

         temp = at[j];
         at[j] = at[j + 1];
         at[j + 1] = temp;

         temp = bt[j];
         bt[j] = bt[j + 1];
         bt[j + 1] = temp;
      }
   }
}

for (i = 0; i < n; i++) {

   if (time < at[i])
      time = at[i];

   time = time + bt[i];
```

```
        ct[i] = time;
        tat[i] = ct[i] - at[i];
        wt[i] = tat[i] - bt[i];

        avg_wt += wt[i];
        avg_tat += tat[i];
    }

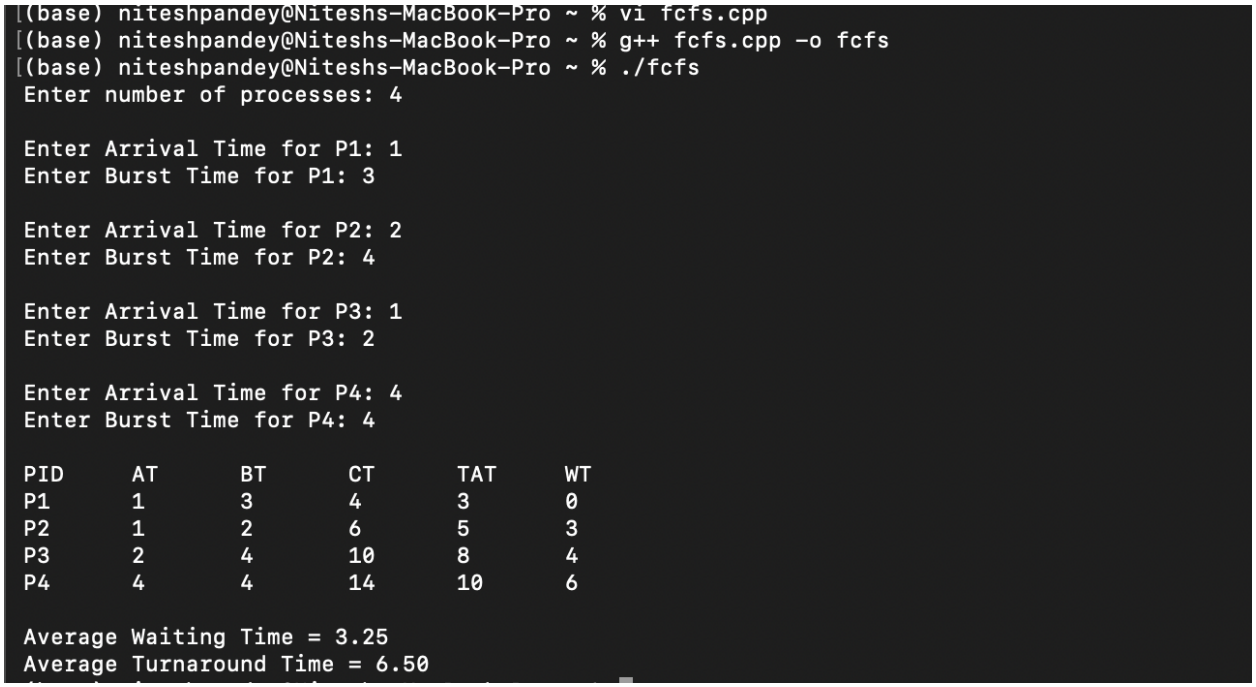
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
            i + 1, at[i], bt[i], ct[i], tat[i], wt[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

Output:



2.SJF

Code:

```
#include <stdio.h>


int main() {
   int n;
   int at[20], bt[20], ct[20], tat[20], wt[20];
   int completed[20] = {0};


   int time = 0;
   int count = 0;
```

```
float avg_wt = 0, avg_tat = 0;

printf("Enter number of processes: ");
scanf("%d", &n);

for (int i = 0; i < n; i++) {
   printf("\nEnter Arrival Time for P%d: ", i + 1);
   scanf("%d", &at[i]);

   printf("Enter Burst Time for P%d: ", i + 1);
   scanf("%d", &bt[i]);
}

while (count < n) {

   int index = -1;
   int shortest = 9999;

   // Find shortest available process
   for (int i = 0; i < n; i++) {

      if (completed[i] == 0 &&
         at[i] <= time &&
         bt[i] < shortest) {
```

```
            shortest = bt[i];
            index = i;
        }
    }

    // No process has arrived yet
    if (index == -1) {
        time++;
        continue;
    }

    // Execute selected process
    time = time + bt[index];

    ct[index] = time;
    tat[index] = ct[index] - at[index];
    wt[index] = tat[index] - bt[index];

    completed[index] = 1;
    count++;

    avg_wt += wt[index];
    avg_tat += tat[index];
}

printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");
```

```
    for (int i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
            i + 1, at[i], bt[i],
            ct[i], tat[i], wt[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

    return 0;
}
```

Output:

```
[(base) niteshpandey@Niteshs-MacBook-Pro ~ % vi sjf.cpp                         ]
[(base) niteshpandey@Niteshs-MacBook-Pro ~ % g++ sjf.cpp -o sjf                 ]
[(base) niteshpandey@Niteshs-MacBook-Pro ~ % ./sjf                              ]
 Enter number of processes: 4

 Enter Arrival Time for P1: 1
 Enter Burst Time for P1: 3

 Enter Arrival Time for P2: 2
 Enter Burst Time for P2: 4

 Enter Arrival Time for P3: 1
 Enter Burst Time for P3: 2

 Enter Arrival Time for P4: 4
 Enter Burst Time for P4: 4

 PID     AT      BT      CT      TAT     WT
 P1      1       3       6       5       2
 P2      2       4       10      8       4
 P3      1       2       3       2       0
 P4      4       4       14      10      6

 Average Waiting Time = 3.00
 Aver      round Time = 6.25
```

3.Round Robin

Code:

```
#include <stdio.h>

int main() {
   int n, quantum;
   int at[20], bt[20], remaining[20];
   int ct[20], tat[20], wt[20];

   int time = 0;
   int completed = 0;

   float avg_wt = 0, avg_tat = 0;

   printf("Enter number of processes: ");
   scanf("%d", &n);

   for (int i = 0; i < n; i++) {
      printf("\nEnter Arrival Time for P%d: ", i + 1);
      scanf("%d", &at[i]);

      printf("Enter Burst Time for P%d: ", i + 1);
      scanf("%d", &bt[i]);

      remaining[i] = bt[i];
   }
```

```
printf("\nEnter Time Quantum: ");
scanf("%d", &quantum);

while (completed < n) {

   int executed = 0;

   for (int i = 0; i < n; i++) {

      if (remaining[i] > 0 && at[i] <= time) {

         executed = 1;

         if (remaining[i] > quantum) {

            time = time + quantum;
            remaining[i] = remaining[i] - quantum;

         } else {

            time = time + remaining[i];
            remaining[i] = 0;

            ct[i] = time;
            tat[i] = ct[i] - at[i];
```

```
            wt[i] = tat[i] - bt[i];

            avg_wt += wt[i];
            avg_tat += tat[i];

            completed++;
          }
        }
      }

      // CPU idle
      if (executed == 0) {
        time++;
      }
    }

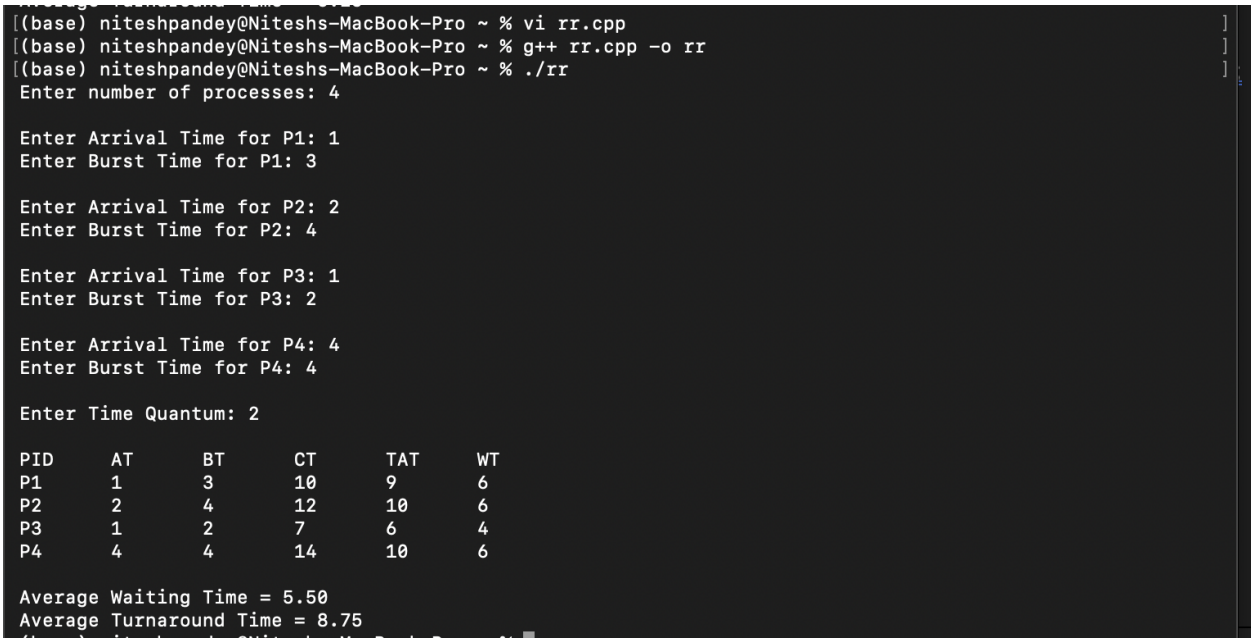
    printf("\nPID\tAT\tBT\tCT\tTAT\tWT\n");

    for (int i = 0; i < n; i++) {
      printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
          i + 1, at[i], bt[i],
          ct[i], tat[i], wt[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);
```

```
    return 0;
}
```

Output:



4.Priority

Code:

```
#include <stdio.h>

int main() {
    int n;

    int at[20], bt[20], priority[20];
    int ct[20], tat[20], wt[20];
```

```
int completed[20] = {0};

int time = 0;
int count = 0;

float avg_wt = 0, avg_tat = 0;

printf("Enter number of processes: ");
scanf("%d", &n);

for (int i = 0; i < n; i++) {

   printf("\nEnter Arrival Time for P%d: ", i + 1);
   scanf("%d", &at[i]);

   printf("Enter Burst Time for P%d: ", i + 1);
   scanf("%d", &bt[i]);

   printf("Enter Priority for P%d: ", i + 1);
   scanf("%d", &priority[i]);
}

while (count < n) {

   int index = -1;
```

```
int highest = 9999;

// Find highest priority available process
for (int i = 0; i < n; i++) {

   if (completed[i] == 0 &&
      at[i] <= time &&
      priority[i] < highest) {

      highest = priority[i];
      index = i;
   }
}

// No process has arrived
if (index == -1) {
   time++;
   continue;
}

// Execute process completely
time = time + bt[index];

ct[index] = time;
tat[index] = ct[index] - at[index];
wt[index] = tat[index] - bt[index];
```

```
    completed[index] = 1;
    count++;

    avg_wt += wt[index];
    avg_tat += tat[index];
}

printf("\nPID\tAT\tBT\tPriority\tCT\tTAT\tWT\n");

for (int i = 0; i < n; i++) {

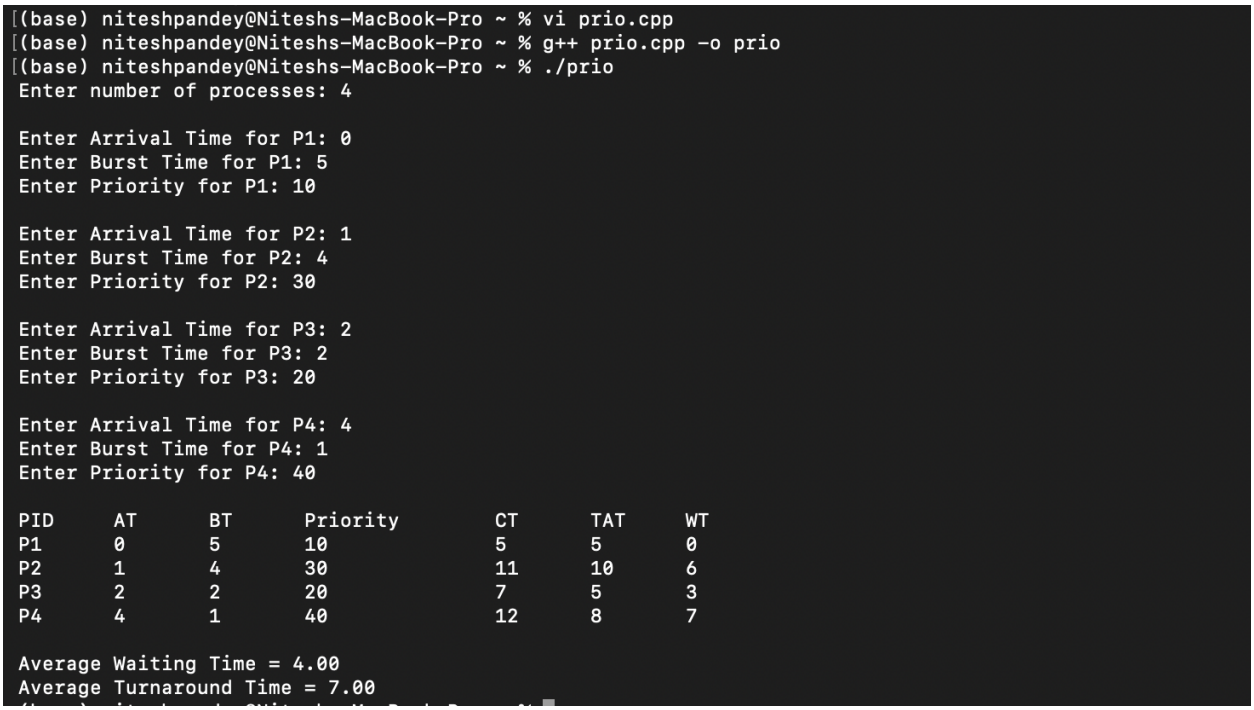
    printf("P%d\t%d\t%d\t%d\t\t%d\t%d\t%d\n",
        i + 1,
        at[i],
        bt[i],
        priority[i],
        ct[i],
        tat[i],
        wt[i]);
}

printf("\nAverage Waiting Time = %.2f", avg_wt / n);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);

return 0;
```

}

Output:


(base) niteshpandey@Niteshs-MacBook-Pro ~ % vi prio.cpp
(base) niteshpandey@Niteshs-MacBook-Pro ~ % g++ prio.cpp -o prio
(base) niteshpandey@Niteshs-MacBook-Pro ~ % ./prio
Enter number of processes: 4
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 5
Enter Priority for P1: 10
Enter Arrival Time for P2: 1
Enter Burst Time for P2: 4
Enter Priority for P2: 30
Enter Arrival Time for P3: 2
Enter Burst Time for P3: 2
Enter Priority for P3: 20
Enter Arrival Time for P4: 4
Enter Burst Time for P4: 1
Enter Priority for P4: 40
PID AT BT Priority CT TAT WT
P1 0 5 10 5 5 0
P2 1 4 30 11 10 6
P3 2 2 20 7 5 3
P4 4 1 40 12 8 7
Average Waiting Time = 4.00
Average Turnaround Time = 7.00